

Final Lab

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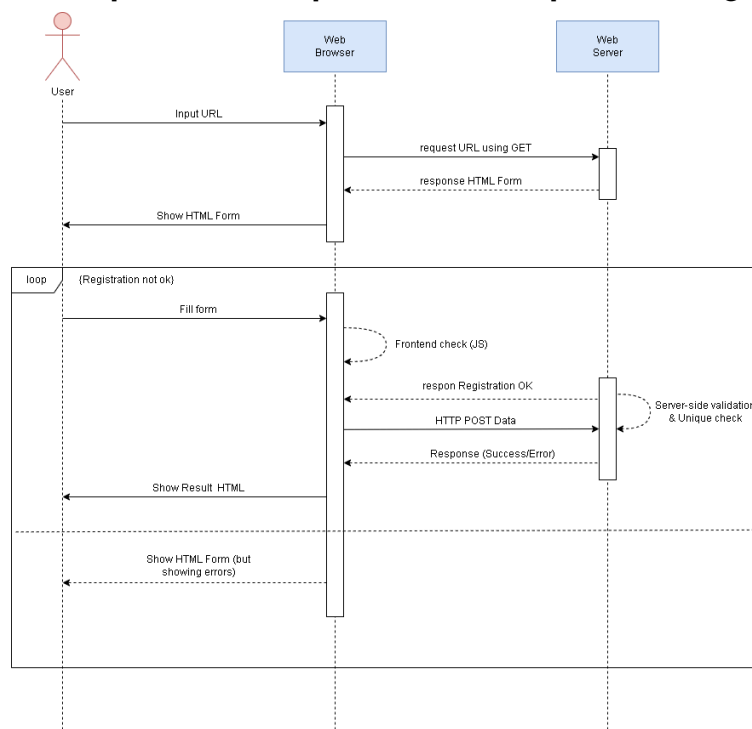
Project Introduction

The idea of our project is a social network designed for sharing and discussing AI-related multimedia content. It functions similarly to Twitter, allowing users to post AI-generated images and engage in community discussions. Social network to share information, experience and pictures about AI.

Core Features:

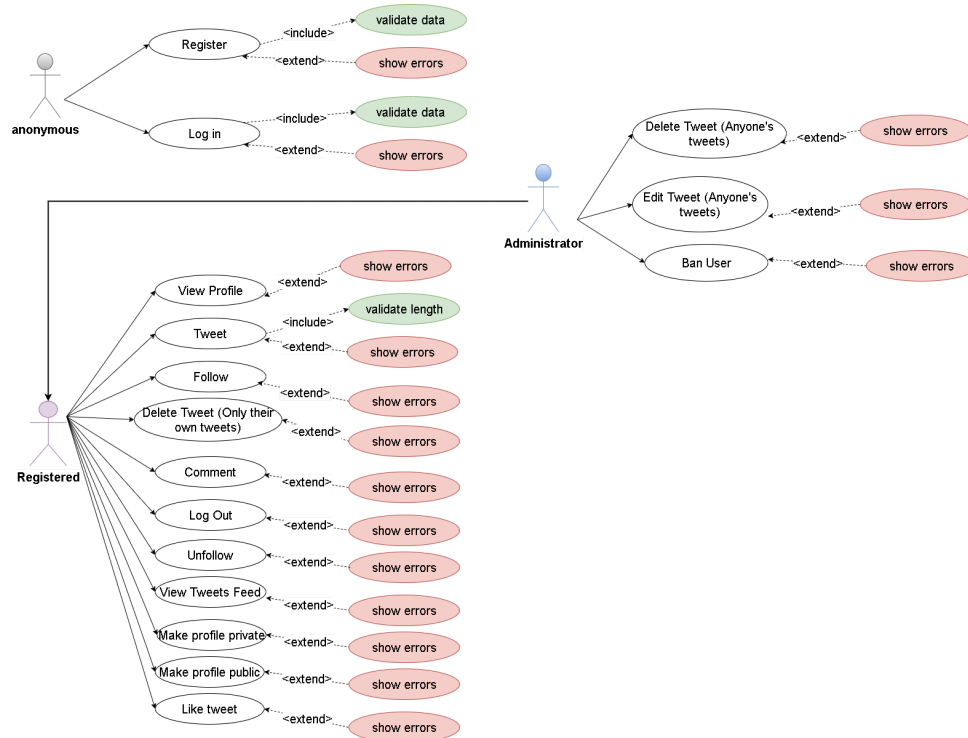
- **Authentication & Authorization:** Secure login, registration, and session management. Role-based access control (Regular Users vs. Admin).
- **Public & Private Timelines:** Registered users can post, edit, delete, and view tweets. Anonymous users can view the global timeline and public profiles in a read-only mode.
- **Hierarchical Interactions:** Users can comment on existing tweets (replies), creating a nested conversation thread.
- **Social Graph:** Users can follow/unfollow others and interact with content through a real-time "Like" system.
- **Dynamic Layout:** The UI adapts its layout (multi-column vs single-column) dynamically based on the context (e.g., viewing the timeline vs viewing a specific user's profile).

Description of the processes (sequence diagram).



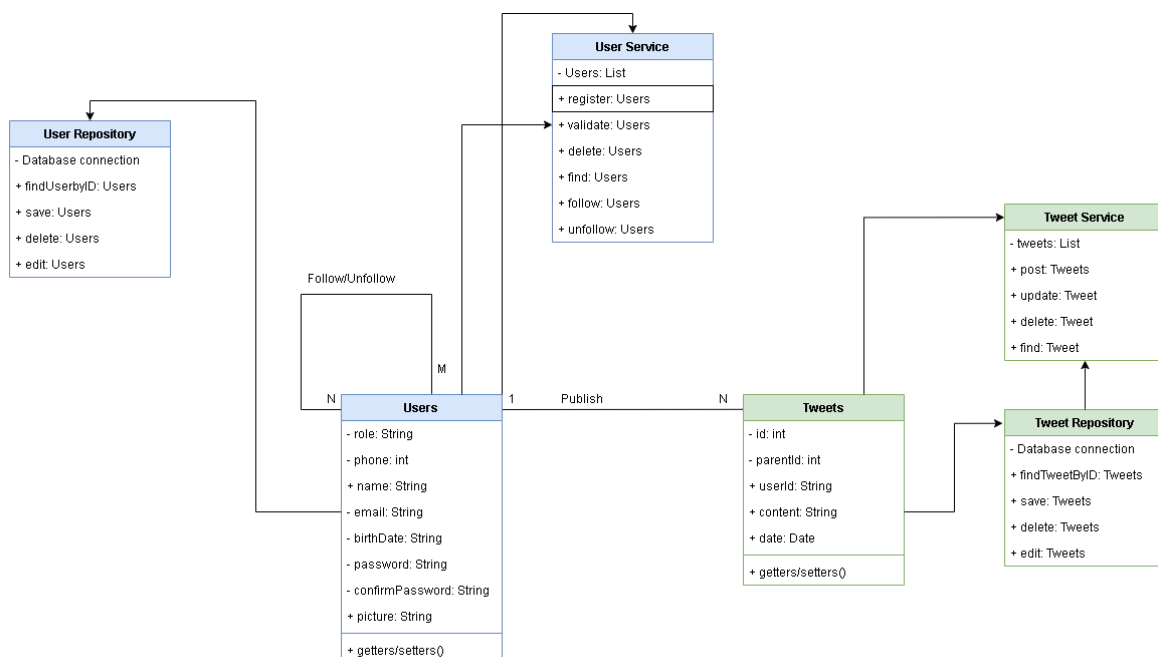
Use case diagram

The following diagram structures user interaction limits. Anonymous clients branch outward into Register or Log in (which include core verification dependencies), while full operational domains are strictly isolated under registered or administrative scopes.

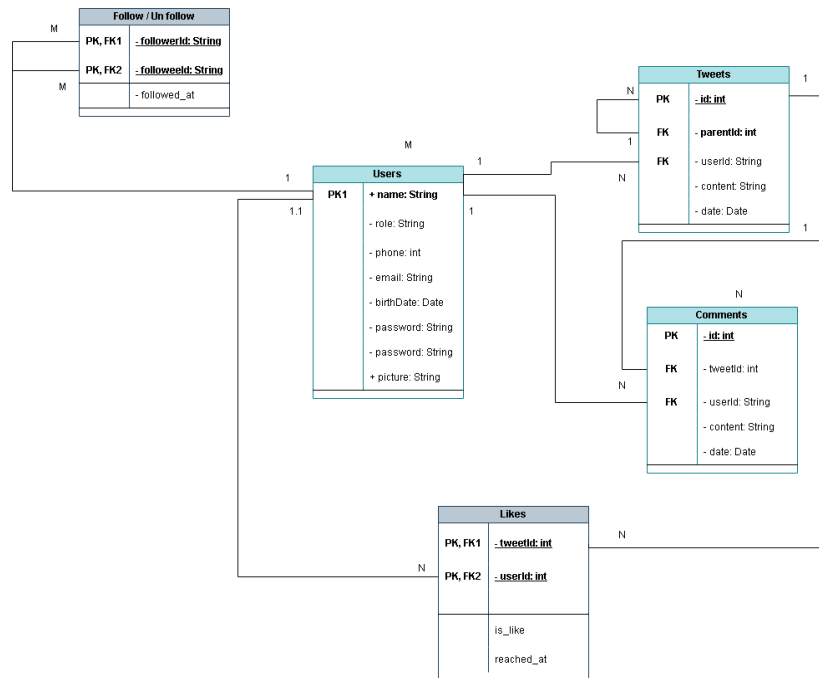


UML class diagram

The structural data engine relies on primary key mapping tables to maintain data integrity across cascades.



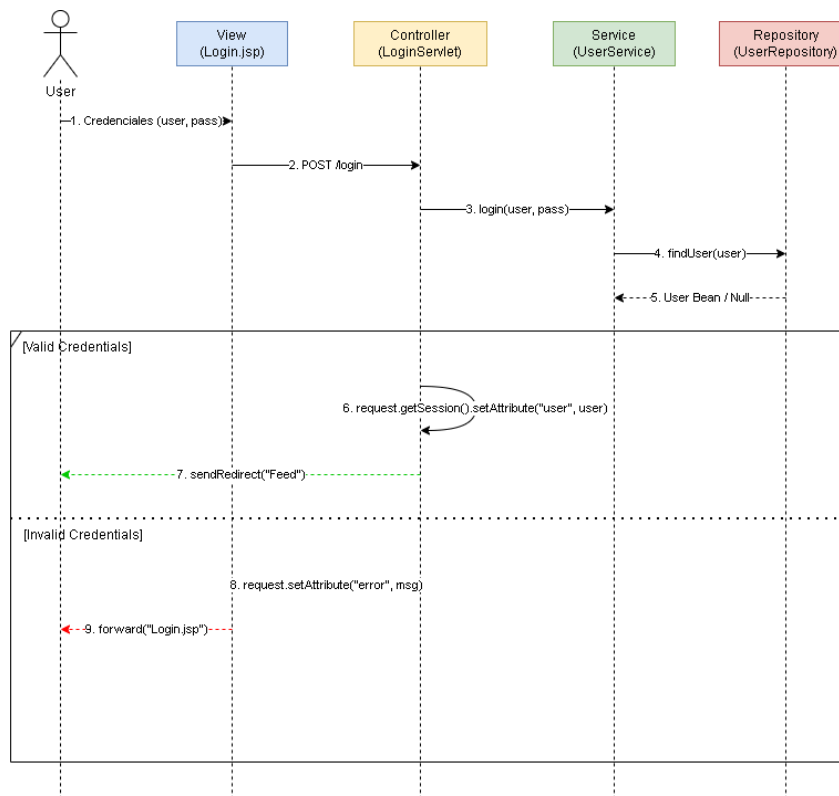
ER Diagram



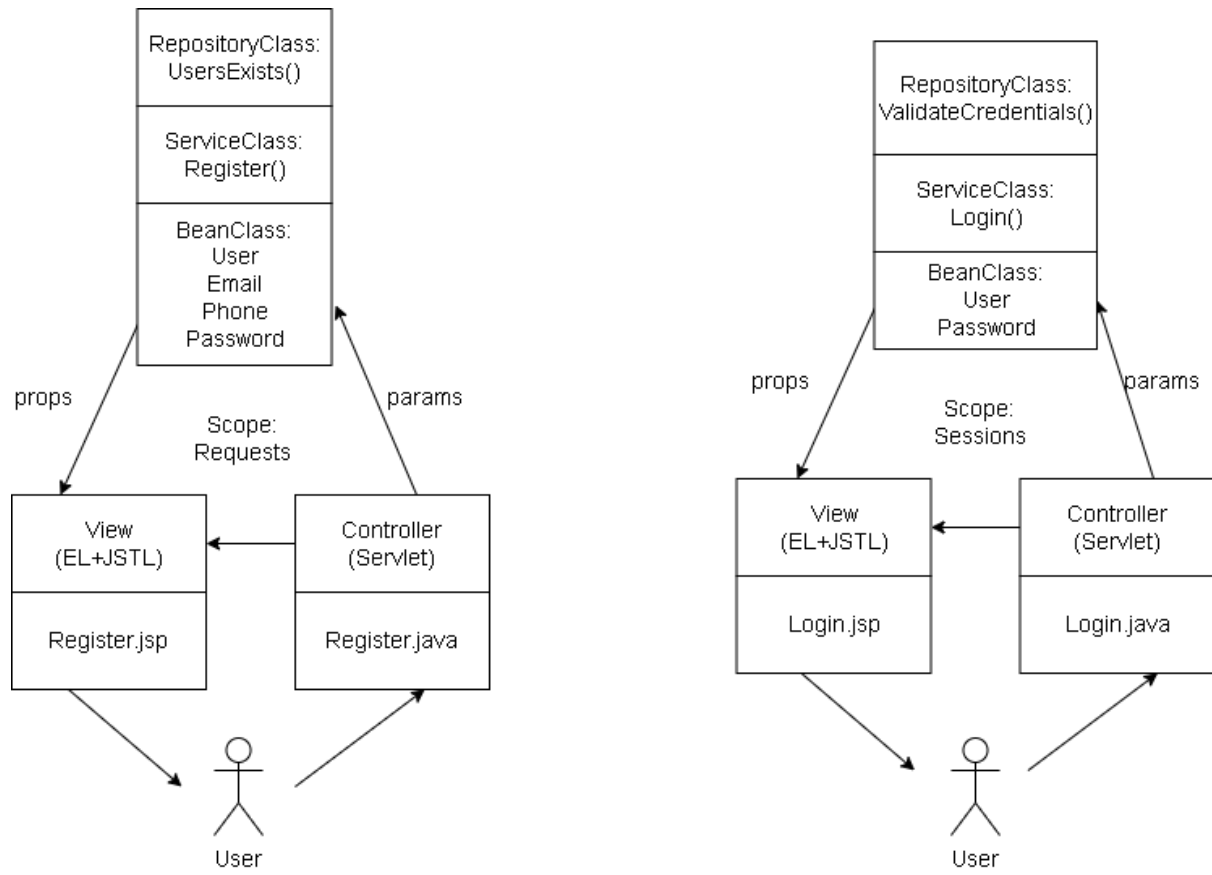
MVC diagram

The registration components utilize dedicated Request scopes to map incoming user state, whereas the login pipeline establishes stateful Session spaces to guard downstream endpoints.

Login functionality using the MVC



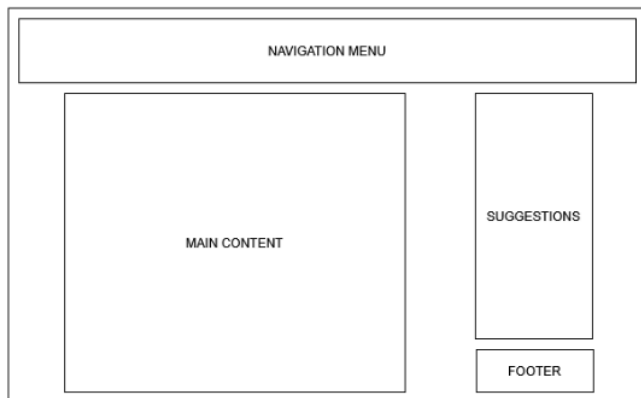
MVC diagram for the login and registration components



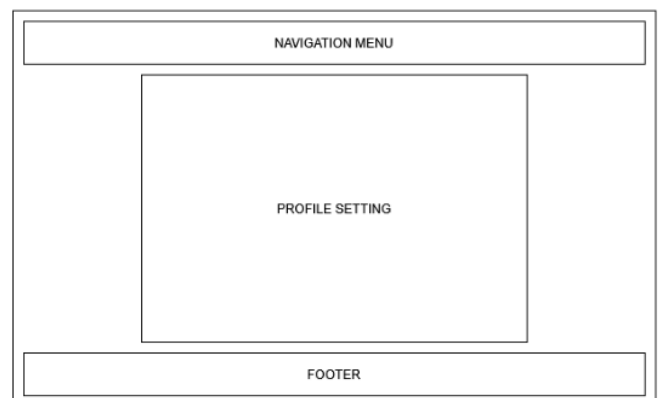
Layout

Layout components are built using structural grids, freeing the views from inline formatting clutter.

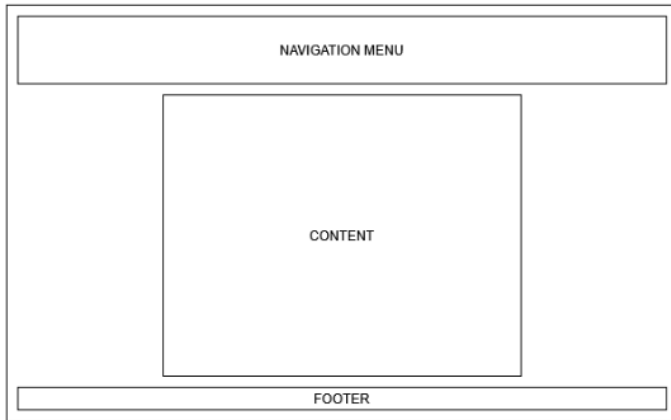
Timeline



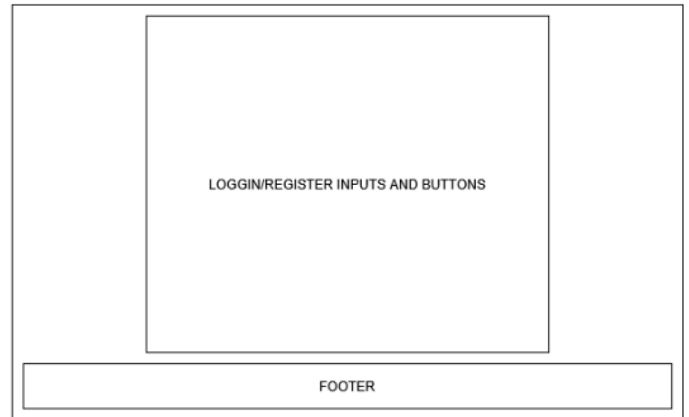
Profile



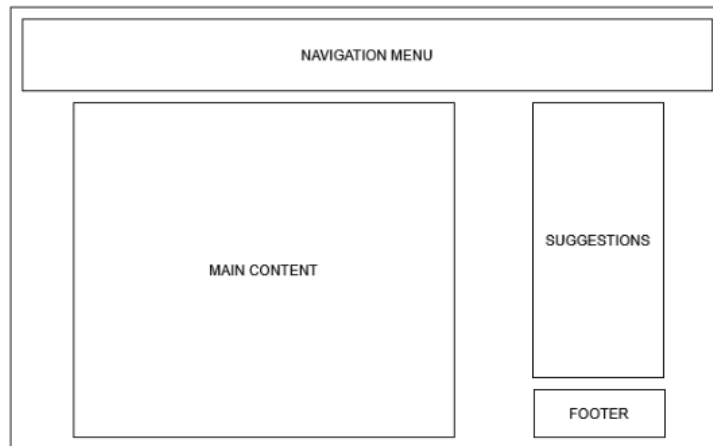
Global Feed



Register/login



Followers/Following



Color Palette

JavaScript

```
:root {
  --bg-color: #0b0e14; /* Deep tech space background */
  --card-bg: #161b22; /* Layered structural surface */
  --text-main: #e6edf3; /* Low-strain text layout */
  --neon: #00d4ff; /* Brand cyan elctric pulse */
  --success: #2ea043; /* Core validation success */
  --error: #ff7b72; /* Structural alert red */
  --border: #30363d; /* Clean dividing edge */
}
```



Iconography

Font Awesome Icons (Basic)



Typography

Main Font Family:

```
font-family: 'Inter', sans-serif;
```

We selected Inter because it is an exceptionally clean geometric sans-serif typeface optimized for screen readability. Its high legibility at small sizes is perfect for rapid data consumption on timelines and dense forms.

Hierarchy:

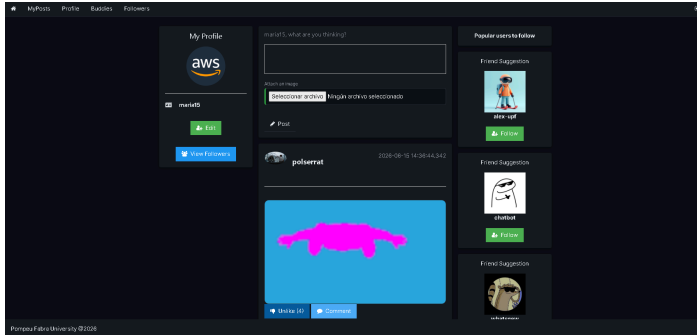
- **Brand Header (h1):** Large (2.5rem), extra bold (800), featuring an integrated glow effect (text-shadow) on the span component to anchor our visual identity.
- **Section Headers (h2):** Crisp (1.1rem), completely capitalized, accompanied by a 4px solid left neon border to catch the eye instantly.
- **Labels & Fields:** Scaled down (0.92rem), bolded (600) with a slight transparency (opacity:

0.92) to maintain structured form layouts.

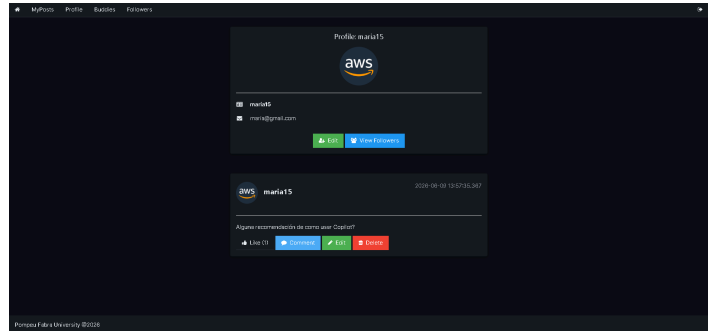
Layout

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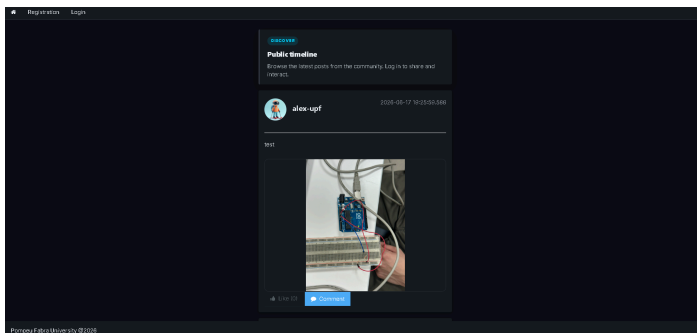
Timeline



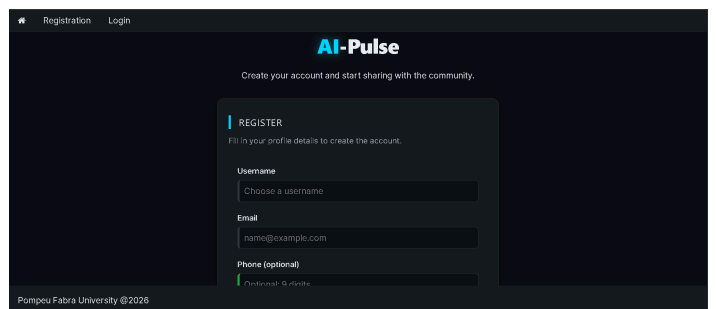
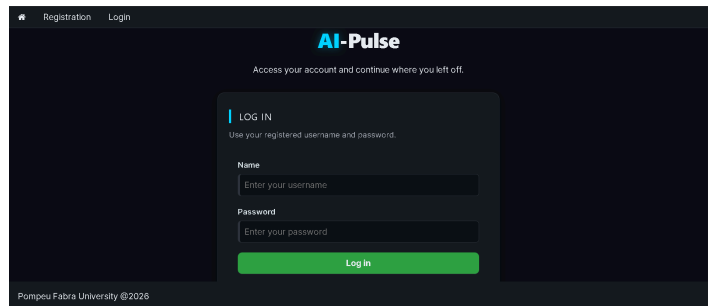
Profile



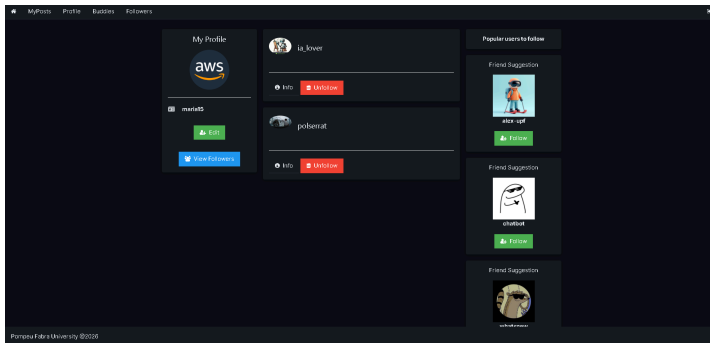
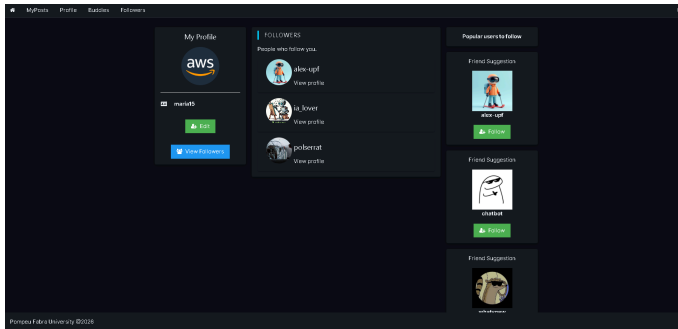
Global Feed



Register/login



Followers/Following



Admin View

